

Ejercicio 1

Dada las ecuaciones diferenciales ordinarias

$$\frac{dy}{dx} = y + x^3$$

Runge-Kutta 4^{to} Orden

$$y(0) = 1; x = [0; 1.5]; h = 0.25$$

$$\frac{dy}{dx} = y + x^3 \quad \begin{matrix} h = 0.25 \\ x = 0 \\ x = 0.25 \\ x = 0.90 \end{matrix}$$

$$y(0) = 1$$

$$h = 0.25$$

$$y(0.25)$$

ITERACIÓN 1

$$x_0 = 0, y_0 = 1$$

$$f(x, y) = y + x^3$$

$$K_1 = f(0, 1) = 1 + 0 = 1$$

$$K_2 = f(0.125, 1 + \frac{0.25}{2}(1))$$

$$K_2 = 1.126953125$$

$$K_3 = f(0.125, 1 + \frac{0.25}{2}(1.126953125))$$

$$K_3 = 1.142822266$$

$$K_4 = f(0.25, 1 + 0.25(1.142822266))$$

$$K_4 = 1.301330566$$

$$y_1 = 1 + \frac{0.25}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

$$y_1 = 1 + \frac{0.25}{6}(1 + 2(1.126953125) + 2(1.142822266) + 1.301330566)$$

$$y_1 \approx 1.28545$$

ITERACIÓN 2

$$x_1 = 0.25 \quad y_1 = 1.28545$$

$$K_1 = 1.28949 + 0.25^3$$

$$K_1 = 1.30108$$

$$K_2 = f(0.375, 1.28545 + 0.125(1.30108))$$

$$K_2 \approx 1.50138$$

$$K_3 \approx 1.52642$$

$$K_4 \approx 1.77294$$

$$y_2 = 1.28545 + \frac{0.25}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

$$y(0.5) \approx 1.665$$

RESPUESTA $y(0.5) \approx 1.66 //$

Ejercicio 2

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} - \left(\frac{n^2}{x^2} - 1\right)y = 0 \quad \text{Para } n=2 \quad \frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} - \left(\frac{4}{x^2} - 1\right)y = 0$$

$$n=2$$

$$y(1)=1$$

$$y'(1)=2$$

$$x \in [1, 3]$$

$$h=0.25$$

$$y_1 = y$$

$$y_2 = \frac{dy}{dx} = y'$$

$$\frac{dy_1}{dx} = y_2$$

$$\frac{dy_2}{dx} = -\frac{1}{x} y_2 + \left(\frac{4}{x^2} - 1\right) y_1$$

$$y_1(1)=1$$

$$y_2(1)=2$$

$$k_{2,1} = f_2\left(x_1 + \frac{h}{2}, y_{1,1} + \frac{h}{2} k_{1,1}, y_{2,1} + \frac{h}{2} k_{2,1}\right) \quad x=1.00, y_1=1.0000, y_2=2.0000$$

$$k_{2,2} = f_2\left(x_1 + \frac{h}{2}, y_{1,1} + \frac{h}{2} k_{1,1}, y_{2,1} + \frac{h}{2} k_{2,1}\right) \quad k_{1,1}$$

$$k_{3,1} = f_3\left(x_1 + \frac{h}{2}, y_{1,1} + \frac{h}{2} k_{1,1}, y_{2,1} + \frac{h}{2} k_{2,1}\right) \quad k_{1,2} = y_2 = 2.0$$

$$k_{3,2} = f_3\left(x_1 + \frac{h}{2}, y_{1,1} + \frac{h}{2} k_{1,1}, y_{2,1} + \frac{h}{2} k_{2,1}\right) \quad k_{1,2} = \frac{1}{2}(2) + \left(\frac{4}{1} - 1\right)(1) = 1.00$$

$$k_{4,1} = f_4\left(x_1 + h, y_{1,1} + h k_{1,1}, y_{2,1} + h k_{2,1}\right) \quad k_{2,1} \text{ en } x=1.125$$

$$k_{4,2} = f_4\left(x_1 + h, y_{1,1} + h k_{1,1}, y_{2,1} + h k_{2,1}\right) \quad y_1 \text{ aux} = 1.2500$$

$$y_{1,1} + h = y_{1,1} + \frac{h}{6} (k_{1,1} + 2k_{1,2} + 2k_{1,3} + k_{1,4}) \quad y_2 = 2.1250$$

$$y_{2,1} + h = y_{2,1} + \frac{h}{6} (k_{2,1} + 2k_{2,2} + 2k_{2,3} + k_{2,4}) \quad k_{2,1} = 2.1605$$

$$y_1(1.25) = 1 + \frac{0.25}{6} (2 + 2(2.125) + 2(2.1015) + 2.2166) \quad k_{2,2} = -0.8217$$

$$= 1.2.6696$$

$$y_2 = 1 + \frac{0.25}{6} (2 + 2(2.125) + 2(2.1015) + 2.2166) \quad k_{3,1} \text{ en } x=1.125$$

$$= 1.5279$$

$$y_2(1.25) = 2 + \frac{0.25}{6} (2 + 2(0.8217) + 2(0.8664) + 0.6063) \quad y_1 \text{ aux} = 1.2656$$

$$= 2.2068$$

$$x=1.25 \quad y_1 = 1.5279 \quad y_2 = 2.2068 \quad y_2 \text{ aux} = 2.1015$$

$$k_{4,1} = 1.56$$

$$k_{4,2} = -1.7733 + 2.3166 = 0.6063$$

Iteración 2 $x = 1.25$ $y_1 = 1.5279$ $y_2 = 2.2068$

$$k_{1,1} = 2.2068 \quad k_{1,2} = -1.6594 + 2.3835 = 0.6181$$

k_2 en $x = 1.375$

$$y_{1,aux} = 1.6279 + 0.125(2.2068) = 1.8937 \quad k_{2,1} = 2.2840$$

$$y_{2,aux} = 2.2840 \quad k_{2,2} = -1.6611 + 2.0120 = 0.3509$$

k_3 en $x = 1.375$

$$y_{1,aux} = 1.8134 \quad k_{3,1} = 2.2506$$

$$y_{2,aux} = 2.2506 \quad k_{3,2} = -1.6368 + 2.0233 = 0.3865$$

k_4 en $x = 1.5$

$$y_{1,aux} = 2.0906 \quad k_{4,1} = 2.2034$$

$$y_{2,aux} = 2.2034 \quad k_{4,2} = -1.5394 + 1.6297 = 0.0901$$

$$y_1(1.5) = \text{suma} = 1.95796, \quad y_1 = 1.9279 + 0.0658 = 2.0937$$

$$y_2(1.5) = \text{suma} = 2.1830, \quad y_2 = 2.2068 + 0.0910 = 2.2977$$

Iteración 3 $x = 1.5$, $y_1 = 2.0937$, $y_2 = 2.2977$

$$k_{1,1} = 2.2977$$

$$k_{1,2} = -1.5318 + 1.6284 = 0.0966$$

k_2 en $x = 1.625$

$$y_{1,aux} = 2.3809 \quad k_{2,1} = 2.5098$$

$$y_{2,aux} = 2.3098 \quad k_{2,2} = -1.4274 + 1.2257 = -0.1997$$

k_3 en $x = 1.625$

$$y_{1,aux} = 2.2824 \quad k_{3,1} = 2.2733$$

$$y_{2,aux} = 2.2733 \quad k_{3,2} = -1.3989 + 1.2265 = -0.1724$$

k_4 en $x = 1.75$

$$y_{1,aux} = 2.0937 + 0.25(2.733) = 2.6650$$

$$y_{2,aux} = 2.2977 + 0.25(-0.1724) = 2.2513$$

$$k_{4,1} = 2.2596$$

$$k_{4,2} = -1.2809 + 0.8250 = -0.4559$$

$$y_1(1.75) = \text{suma} = 1.97289, \quad y_1 = 2.0937 + 0.5766 = 2.6653$$

$$y_2(1.75) = \text{suma} = -1.1171, \quad y_2 = 2.2977 - 0.0969 = 2.2513$$

Iteración 5 $x = 2.0$ $y_1 = 3.2073$ $y_2 = 2.0630$

$$k_{1,1} = 2.0630$$

$$k_{1,2} = -0.5(2.0630) + 0 = -1.0315$$

$$k_2 \text{ en } x = 2.125$$

$$y_{1,aux} = 3.4654 \quad | \quad k_{2,1} = 1.9341 = -0.1142$$

$$y_{2,aux} = 1.9341 \quad | \quad k_{2,2} = -0.9101 - 0.3958 = -1.3060$$

$$k_3 \text{ en } x = 2.125$$

$$y_{1,aux} = 3.4443 \quad | \quad k_{3,1} = 1.8998$$

$$y_{2,aux} = 1.8998 \quad | \quad k_{3,2} = -0.8938 - 0.3938 = -1.2876$$

$$k_4 \text{ en } x = 2.25$$

$$y_{1,aux} = 3.6826 \quad | \quad k_{4,1} = 1.7411 = -0.2099$$

$$y_{2,aux} = 1.7411 \quad | \quad k_{4,2} = -0.7737 - 0.7728 = -1.5465$$

$$y_1(2.25) = y_{aux} = 1.7411, y_1 = 3.2073 + 0.478 = 3.6853$$

$$y_2(2.25) = y_{aux} = -1.5465, y_2 = 2.0630 = -1.7394$$

Iteración 6 $x = 2.25$ $y_1 = 3.6853$ $y_2 = 1.7394$

$$k_{1,1} = 1.7394$$

$$k_{1,2} = -0.7731 - 0.7739 = -1.5470$$

$$k_2 \text{ en } x = 2.375$$

$$y_{1,aux} = 3.9030 \quad | \quad k_{2,1} = 1.5461 = -0.2909$$

$$y_{2,aux} = 1.5461 \quad | \quad k_{2,2} = -0.6510 - 1.1353 = -1.7863$$

$$k_3 \text{ en } x = 2.375$$

$$y_1 = 3.8788 \quad | \quad k_{3,1} = 1.5262$$

$$y_2 = 1.5262 \quad | \quad k_{3,2} = -1.7866$$

Iteración 7 $x=2.6, y_1=4.0611, y_2=1.2863$

$$K_{1,1} = 1.2963$$

$$K_{1,2} = -1.9827$$

Cálculo de f_2 en $x=2.629$

$$y_1 = 4.2293 \quad K_{2,1} = 4.0487 = -0.9199$$

$$y_2 = 1.0489 \quad K_{2,2} = -2.2738$$

$$K_{2,1} = -0.9199$$

$$K_{2,2} = -2.2738$$

K_3 en $x=2.629$

$$y_1 = 4.2981$$

$$y_2 = 1.0297$$

$$K_{3,1} = 1.0297$$

$$K_{3,2} = -2.1917$$

K_4 en $x=2.79$

$$y_1 = 4.3235$$

$$y_2 = 0.7584$$

$$K_{4,1} = 0.7584 = -0.9711$$

$$K_{4,2} = -2.3127$$

Iteración 8 $x=2.75, y_1=4.3257, y_2=0.7569$

$$K_{1,1} = 0.7569$$

$$K_{1,2} = -2.3129$$

$$K_{2,1} = 4.4203$$

$$K_{2,2} = 0.9678$$

$$K_{2,1} = 0.9678$$

$$K_{2,2} = -0.1627 - 2.2813 = -2.4440$$

K_3 en $x=2.875$

$$y_1 = 4.3257 + 0.125(0.4618) = 4.3892$$

$$y_2 = 0.7569 + 0.125(-2.4440) = 0.9519$$

$$K_{3,1} = 0.9519$$

$$K_{3,2} = -0.3978(0.9519) + (-0.3261)(4.3892) = -2.4199$$

K_4 en $x=3.0$

$$y_1 = 4.3257 + 0.25(0.4914) = 4.4386$$

$$y_2 = 0.7569 + 0.25(-2.4191) = 0.1521$$

$$K_{4,1} = 0.1521$$

$$K_{4,2} = -0.5393(0.1521) + (-0.5556)(4.4386) = -0.0907 - 2.4659 = -2.5566$$

$$y_1(3,0) = 4.3297 + 0.0416667(0.7969 + 0.9356 + 0.9028 + 0.1521)$$

$$\text{suma} = 2.7744, y_1 = 4.3297 + 0.1195 = 4.4492$$

$$y_2(3,0) = 0.7969 + 0.0416667(-2.3129 - 4.8880 - 4.8388 - 2.9166)$$

$$\text{suma} = -19.9563, y_2 = 0.7969 - 0.2063 = 0.1906$$

$$y(3,0) \approx 4.4402$$

$$\text{RESPUESTA} = 4.4 //$$